



PELEAS – Air Quality Predictor

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Problem

- Social Problems
- Engineering Problems



Social Problems

- Vulnerable population is at risk
- No way to plan beforehand outdoor activities
- Reactive responses to danger instead of proactive planning



Engineering Problems

- Data quality and sparsity
- Time-series forecasting complexity
- Evaluation across horizons*



Solution



72-hour air quality prediction.

Plan your day based on the air quality
just like you do with the weather!



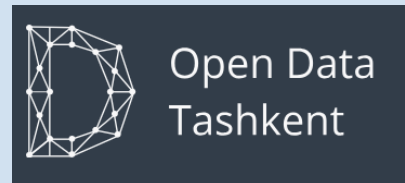
Methodology & Implementation

- **Part 1 – Data**
- **Part 2 - AI**



Data

Дата	Время	Станция	AQI	Влажность	Температура	Осадки	PM2.5
2025-12-09	23:00:00	детский сад №423	88,64377682	99,9	5,2	2,2	30
2025-12-09	23:00:00	Детский сад 557, Тузель-1	57,09871245	99,9	5,6		15
2025-12-09	23:00:00	Бектемирский район детский сад №578		99,9	6	1,6	7568
2025-12-09	23:00:00	Янгихаятский район №школа 333		100	0		7
2025-12-09	23:00:00	Парк Локомотив	99,15879828	99,9		1	35
2025-12-09	23:00:00	Парк Гафура Гулама	53,94420601	99,9		0	13,5
2025-12-09	23:00:00	Алмазарский район Дерматология	50,78969957	100		2,6	12
2025-12-09	22:00:00	Бектемирский район детский сад №578	69,7167382	99,9	5,85	0,4	21



Data

10 Station:

1. Almazar
2. Bektemir
3. ...

7 Variables:

1. AQI
2. Humidity
3. PM2.5
4. Temperature
5. U10
6. V10
7. *Wind Speed*

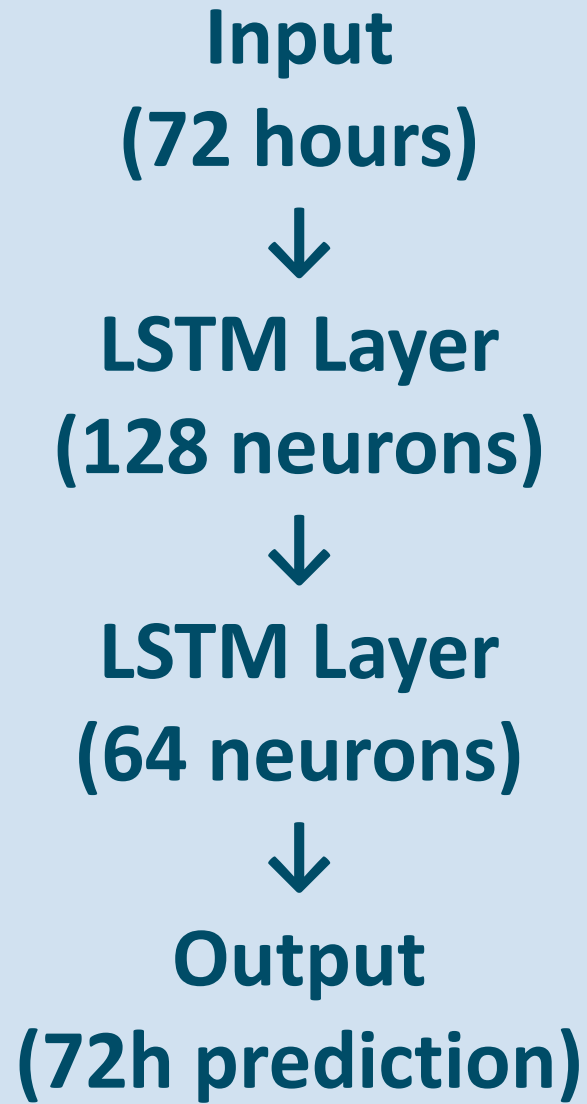


Data

Data Split:
80% training
20% testing



AI

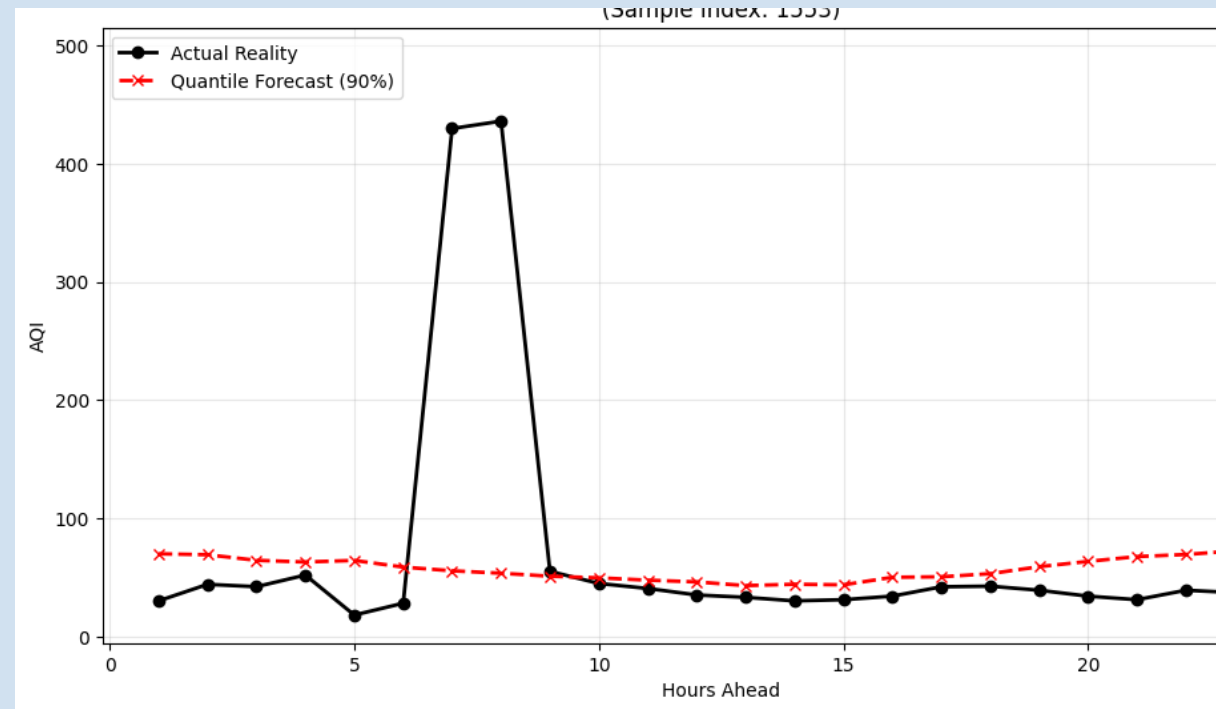


Dropout rate 20%

{ LSTM Layer – extracts long-term trends.
LSTM Layer – compresses features

Challenge 1

Our initial model XGBoost + Decision Tree was "too safe," predicting flat lines and under-estimating pollution spikes.

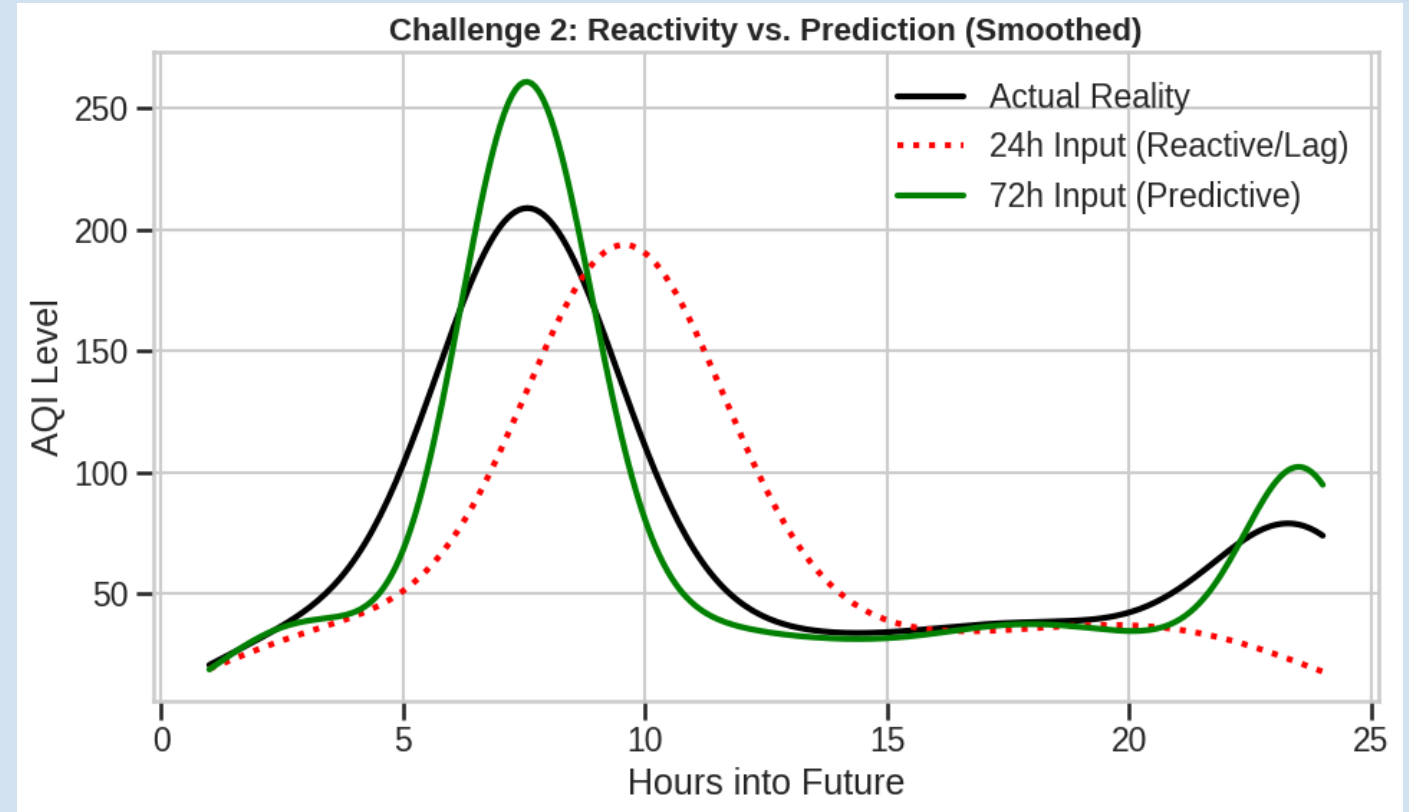


Challenge 1 - Solution

Switched to LSTM to better capture the sequence and momentum of pollution rising.

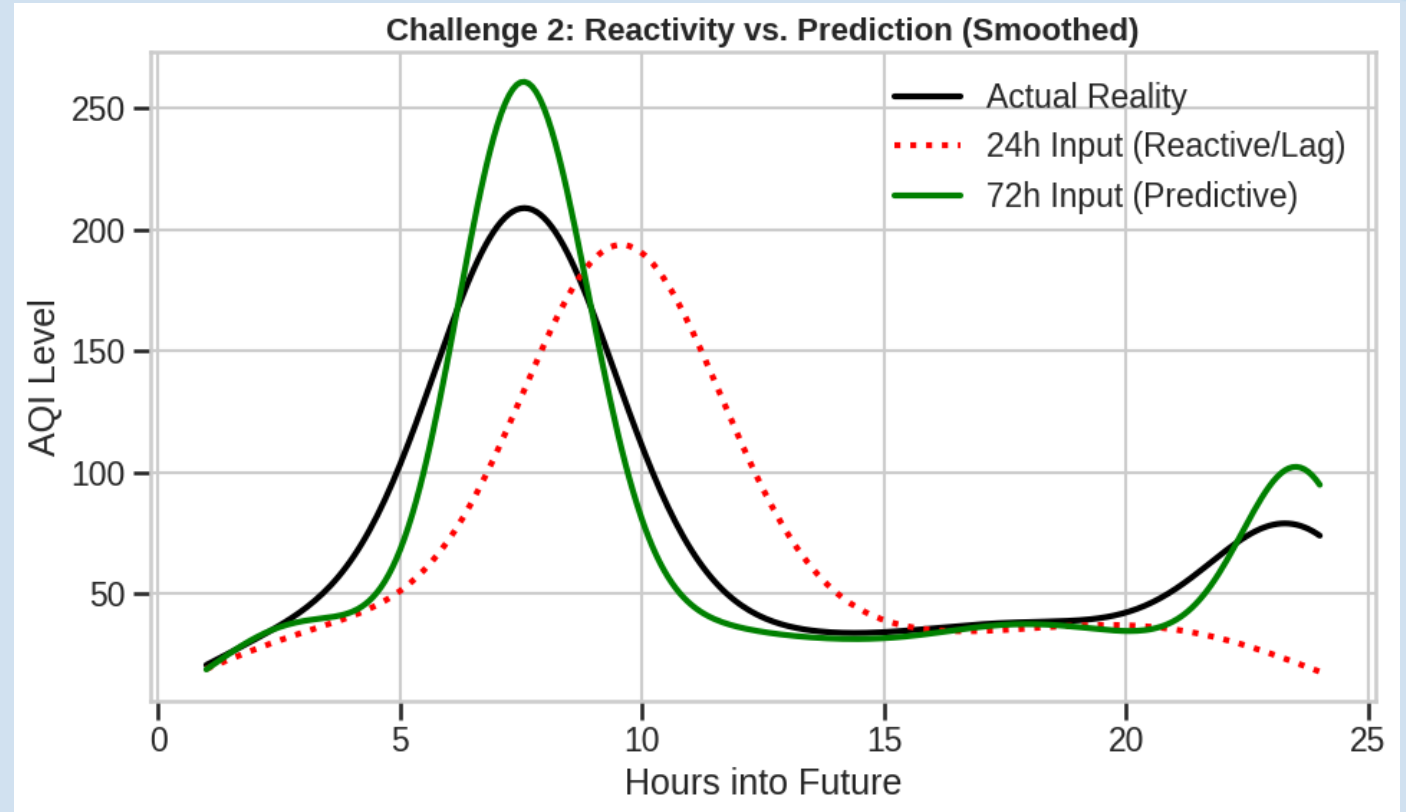
Challenge 2

The model was reacting right, but late



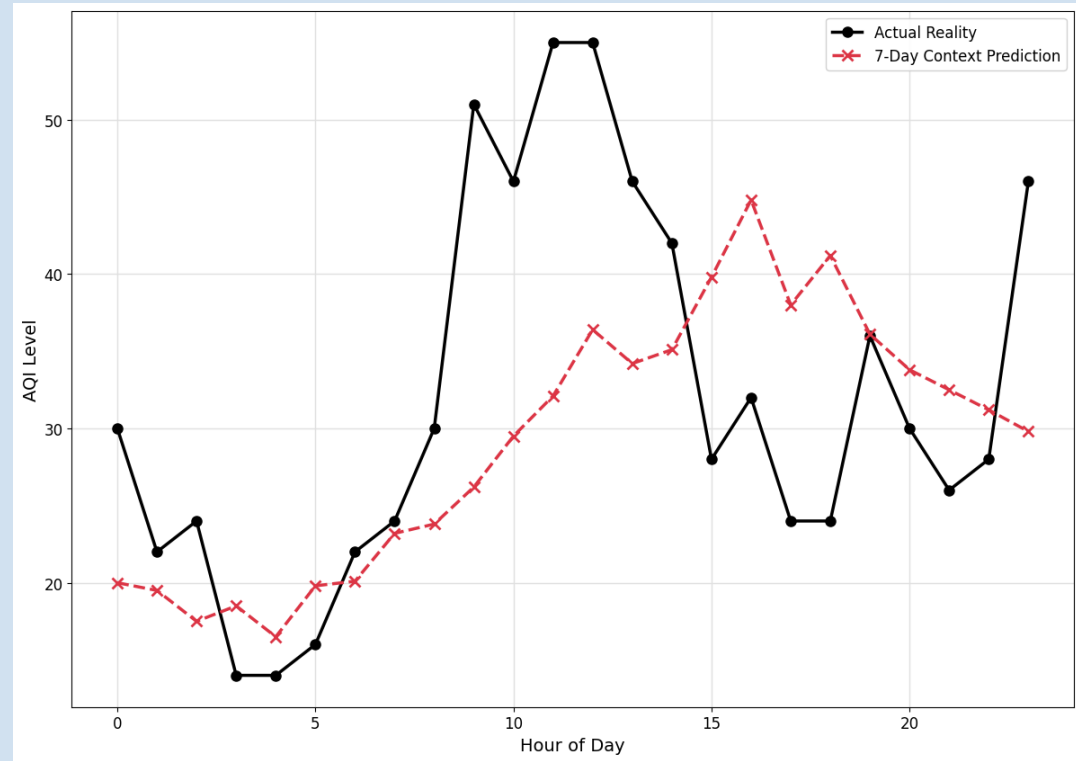
Challenge 2 - Solution

We increased the input context from 24 hours to 72 hours. This allowed the model to see the "buildup" phase before the spike occurred



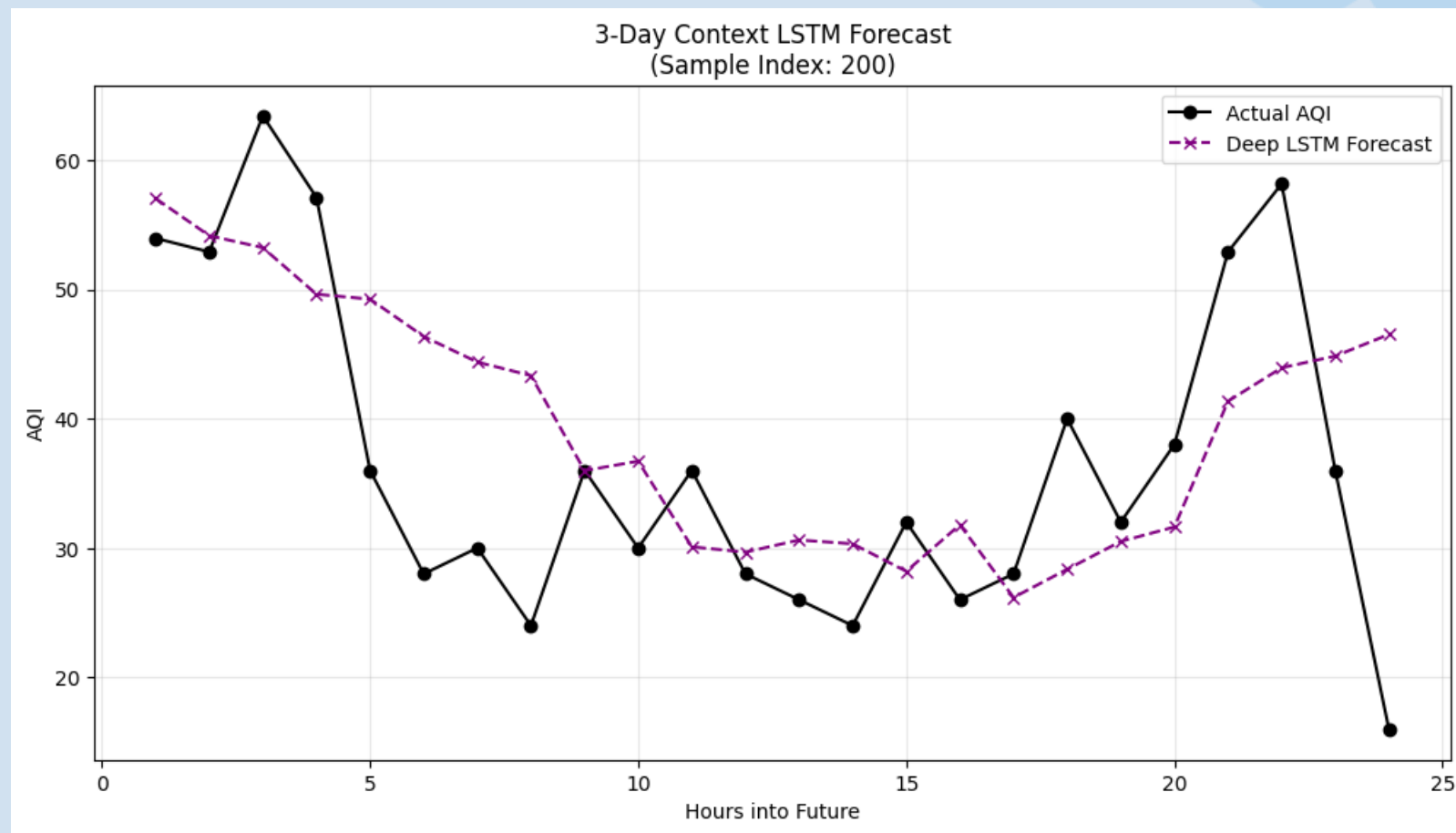
Challenge 3

We reached the
bottleneck, no
improvement past MAE
22

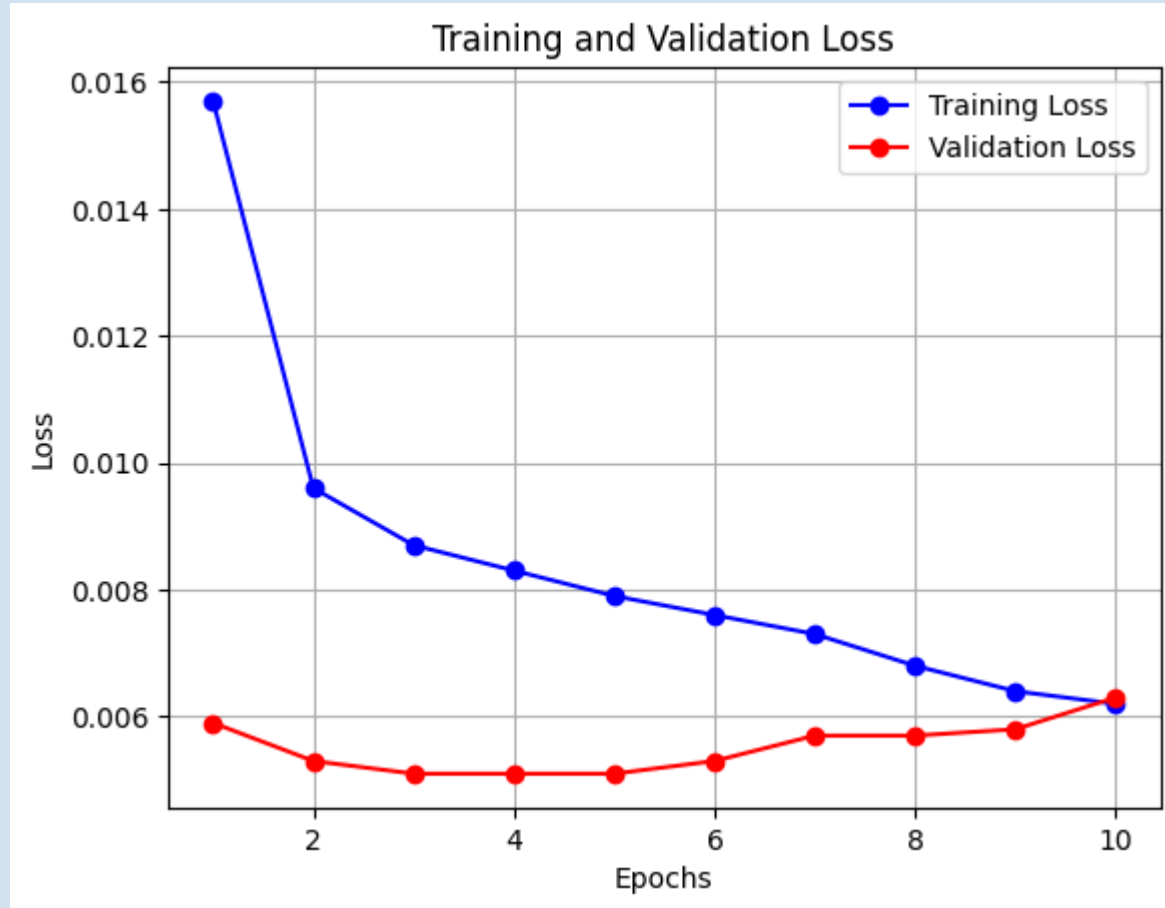


Challenge 3 - Solution

We added wind direction and speed data, achieved MAE 19.4



Experiment & Results



10 epochs

Experiments & Results

MAE on regular days: ~19

MAE on days with sudden spikes* of AQI: ~20-35

***sudden spikes = up/down for more than 200 pts**

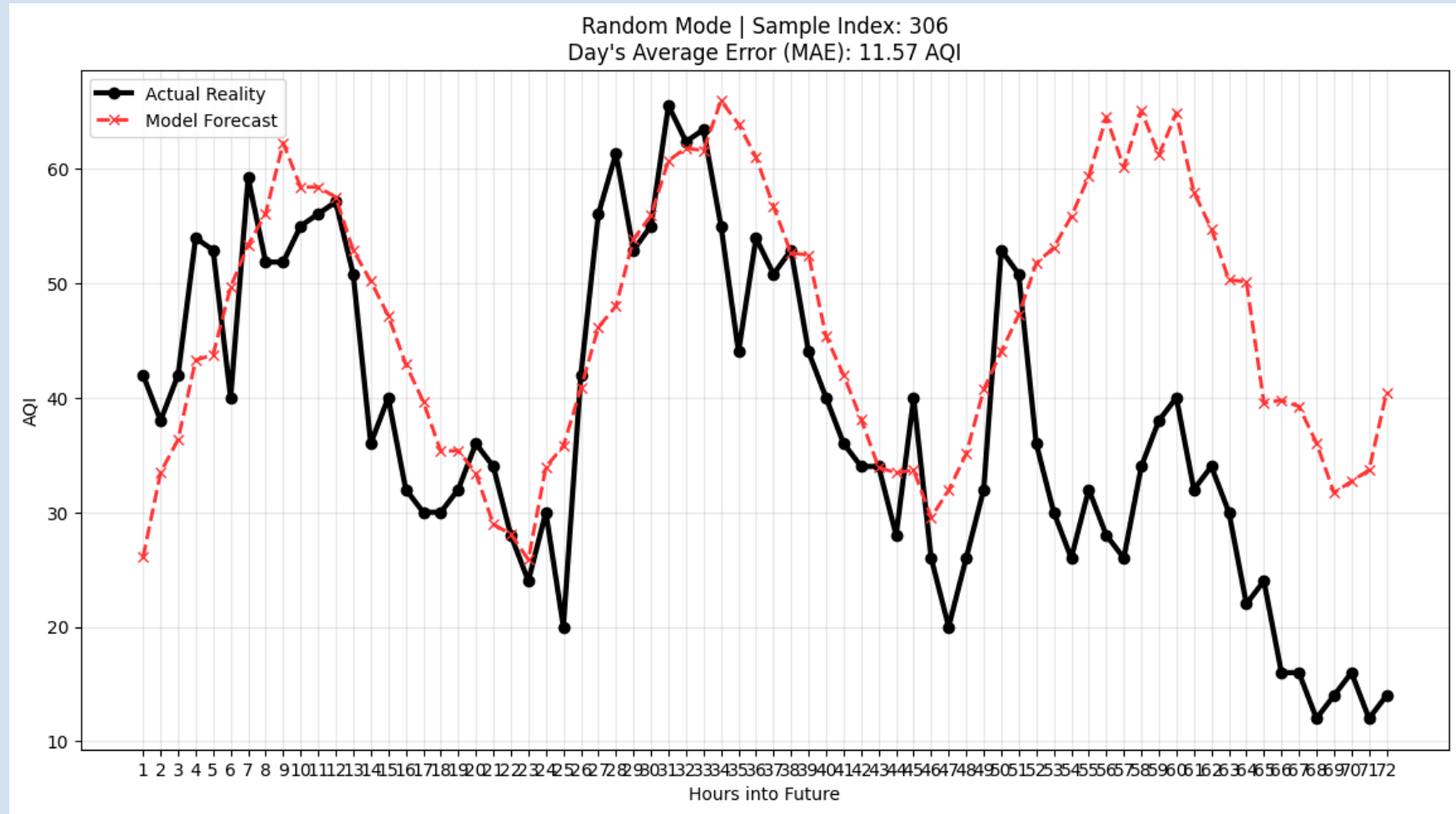
Experiments & Results

1st Hour RMSE:
29.0658

24th Hour RMSE:
34.3596

48th Hour RMSE:
35.1374

72nd Hour RMSE:
35.4608



Conclusion

Achievement: Air pollution prediction.

Importance: Warns people about possible health harms in the near 72h future

Takeaway:

“Before tuning the model fix with data”

Andrew Ng



Links

<https://github.com/JavohirX/peleas>